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| <b>Topic 6A: Introduction to organic chemistry</b>                                                                                                                                                                                            |
| 1. know that a hydrocarbon is a compound of hydrogen and carbon only                                                                                                                                                                          |
| 2. be able to represent organic molecules using empirical formulae, molecular formulae, general formulae, structural formulae, displayed formulae and skeletal formulae                                                                       |
| 3. know what is meant by the terms 'homologous series' and 'functional group'                                                                                                                                                                 |
| 4. be able to name compounds relevant to this specification using the rules of International Union of Pure and Applied Chemistry (IUPAC) nomenclature<br><i>Students will be expected to know prefixes for compounds up to C<sub>10</sub></i> |
| 5. be able to classify reactions as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation                                                                                                                   |
| 6. understand the term 'structural isomerism' and determine the possible structural, displayed and skeletal formulae of an organic molecule, given its molecular formula                                                                      |
| 7. understand the term 'stereoisomerism', as illustrated by <i>E/Z</i> isomerism (including <i>cis-trans</i> isomerism where two of the substituent groups are the same)                                                                      |